

Demand Planning

Using Artificial Intelligence & Machine Learning

Case Study: Siemens Mobility

Supply Chain Analytics | M.Sc. Data Science – Business Analytics

Agenda

01 Business Context

Rail infrastructure demand environment

02 Demand Planning

What it is and why it matters

03 Traditional Forecasting

Methods, strengths, and limitations

04 Bullwhip Effect

The Bullwhip Effect & information asymmetry

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Expanding beyond historical patterns

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07 Feature Engineering

Internal and external signal integration

08 S&OP Integration

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Uncertainty management through AI augmentation

Business Context: Rail Infrastructure Demand

Siemens Mobility — Business Segments

-  Rolling Stock
-  Rail Infrastructure
-  Signaling Systems
-  Electrification
-  Lifecycle Services

Structural Demand Characteristics

-  **Project-Based Demand**
No steady-state consumption pattern
-  **Long Lead Times**
18–36 months for major components
-  **Government Dependency**
Public funding cycles drive volumes
-  **High Unit Value**
Customised, Based on requirement volume components
-  **Structural Uncertainty**
Cannot rely on historical mean alone

What is Demand Planning?



The Demand-to-Delivery Chain

Consequences of Forecast Error



Excess Inventory

Higher working capital costs



Backlog Accumulation

Unmet demand & delivery delays



Service-Level Risk

Customer satisfaction impact



Supplier Instability

Reactive procurement costs



The Bullwhip Effect

Small changes in customer demand create amplified fluctuations upstream in the supply chain.

The Bullwhip Effect is famously illustrated by the 'Beer Game,' a supply chain simulation developed at MIT.



The Beer Game:

Supply Chain Instability & the Bullwhip Effect



← Demand signal amplifies upstream (Bullwhip Effect)

Round 1 — No Demand Visibility

- High backlog accumulation
- Large inventory swings
- Excessive ordering amplification
- High total supply chain cost

Round 2 — Demand Shared

- Stable, low backlog levels
- Smooth inventory transitions
- Reduced ordering overreaction
- Lower total supply chain cost

Key Insight: Better information flow reduces the Bullwhip Effect - this is why AI-driven forecasting matters.

Traditional Forecasting Methods

ARIMA

AutoRegressive Integrated Moving Average

How: Uses past demand values and error terms

Best when: Stable, linear time-series data

Exp. Smoothing

Exponential Smoothing (ETS)

How: Weighted average of recent observations

Best when: Short-term forecasts with trend/seasonality

Planner Judgment

Expert-Based Judgment

How: Planner knowledge of market conditions

Best when: Low-data environments, qualitative signals

Safety Stock Buffers

Statistical Safety Stock

How: Buffer inventory = $f(\text{demand variability})$

Best when: Hedge against forecast uncertainty

⚠ Core Limitation of Traditional Methods

All traditional methods rely primarily on historical demand patterns and assume relative stability over time. They fail when demand is driven by structural external factors ; such as government budget cycles, infrastructure investment programmes, or macroeconomic policy shifts ; which are common in project-based rail environments.

Machine Learning Techniques for Demand Forecasting

ARIMA

Baseline

- Uses past demand + lagged errors
- Linear time-series assumption
- Best: stationary, seasonal data
- Benchmark for ML comparison

Best for: Baseline model

Random Forest

Ensemble Trees

- Many decision trees combined
- Captures nonlinear interactions
- Robust to noisy structured data
- Interpretable feature importances

Best for: Complex structured data

XGBoost

Gradient Boosting

- Sequential error correction
- Strong on tabular ERP data
- High predictive accuracy
- Widely used in industry

Best for: ERP / backlog features

LSTM

Deep Learning

- Neural network with memory cells
- Learns long-term sequences
- Captures temporal dependencies
- Requires more training data

Best for: Long-range patterns

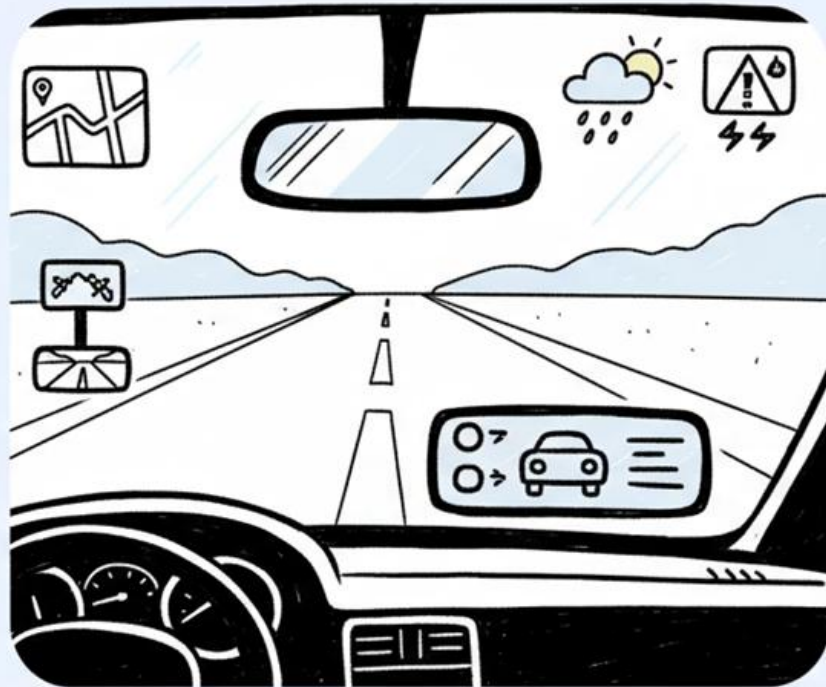
How AI Expands Demand Planning

Dimension	Traditional Methods	AI / ML Approach
Data inputs	Historical demand only	Historical + external + backlog signals
Relationship type	Linear, additive	Nonlinear, interaction effects
Structural shifts	Poor detection	Improved via feature engineering
Output type	Point estimate	Probabilistic forecast distribution
Bias handling	Manual correction	Data-driven bias reduction
Learning over time	Static parameters	Model retraining on new data

TRADITIONAL

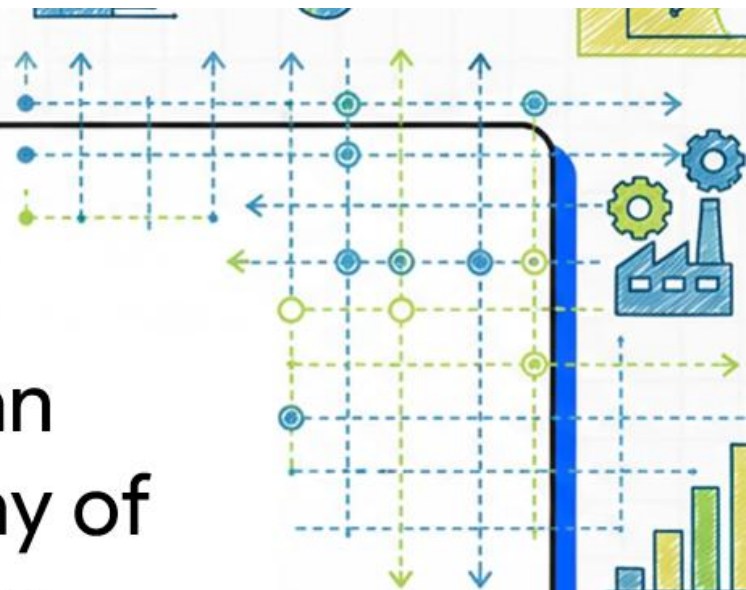


MACHINE LEARNING



“

So, Does this mean
the Traditional way of
Forecasting has no
impact at all?



Feature Engineering: What Drives Rail Demand?

Internal Features

Backlog Size

Volume of undelivered orders on hand

Order Frequency

Rate of new project order arrivals

Lead-Time Variability

Fluctuations in supplier delivery time

Production Capacity

Current available manufacturing capacity

Historical Demand

Lagged demand for ARIMA-style patterns

External Features

Infrastructure Spending Index

Government capex commitments in rail

Policy & Regulatory Signals

Tender announcements, budget cycles

Macroeconomic Indicators

GDP growth, construction indices

Commodity Prices

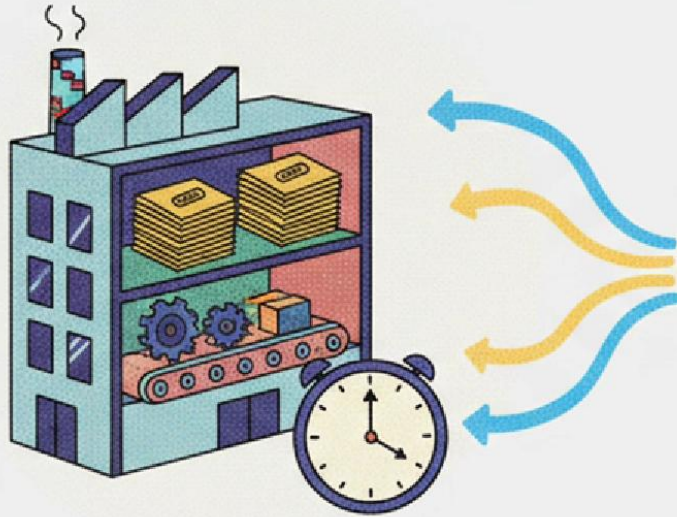
Steel, copper — input cost signals

Customer Project Milestones

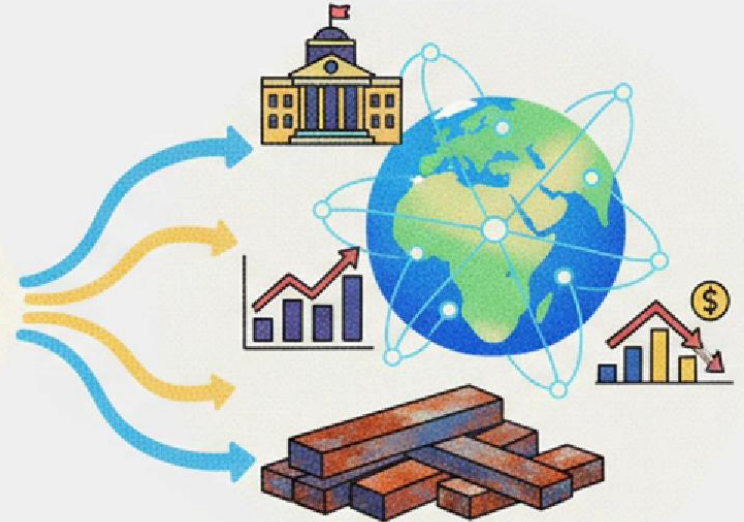
Known delivery windows from contracts

ML model quality depends on feature quality — combining internal and external signals is essential in rail.

Internal Features



External Features



Why AI Improves Demand Planning

01

Captures Nonlinear Relationships

Models complex interactions between backlog, lead time, and economic signals that linear models miss.

02

Multi-Signal Integration

Combines internal ERP data with external macroeconomic and policy indicators in a single model.

03

Reduces Forecast Bias

Data-driven correction of systematic over- or under-forecasting tendencies.

04

Detects Structural Shifts

Responds faster to changes in government spending cycles or project award patterns.

05

Probabilistic Forecasting

Provides confidence intervals rather than point estimates - enabling risk-informed planning.

06

Uncertainty Management

Goal is not perfect accuracy - it is structured, defensible management of forecast uncertainty.

Integration into S&OP and Governance



1. Data Layer

Gathers internal ERP data and external economic indicators.

2. Model Layer

ML models generate probabilistic forecasts.



3. Decision Layer

Planners validate insights for S&OP meetings.



Governance Principle: AI functions as a decision-support tool, not an autonomous system. Planners retain forecast ownership and accountability. All model outputs must be interpretable and explainable to gain planner trust.

Performance Metrics



Forecast Bias



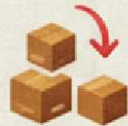
MAPE (Accuracy)



Service Level %



Working Capital



Backlog Reduction

Implementation Considerations

1. Data Quality & Integration

- ERP data often incomplete or siloed
- External data sources need curation
- Data governance requires early investment

2. Change Management

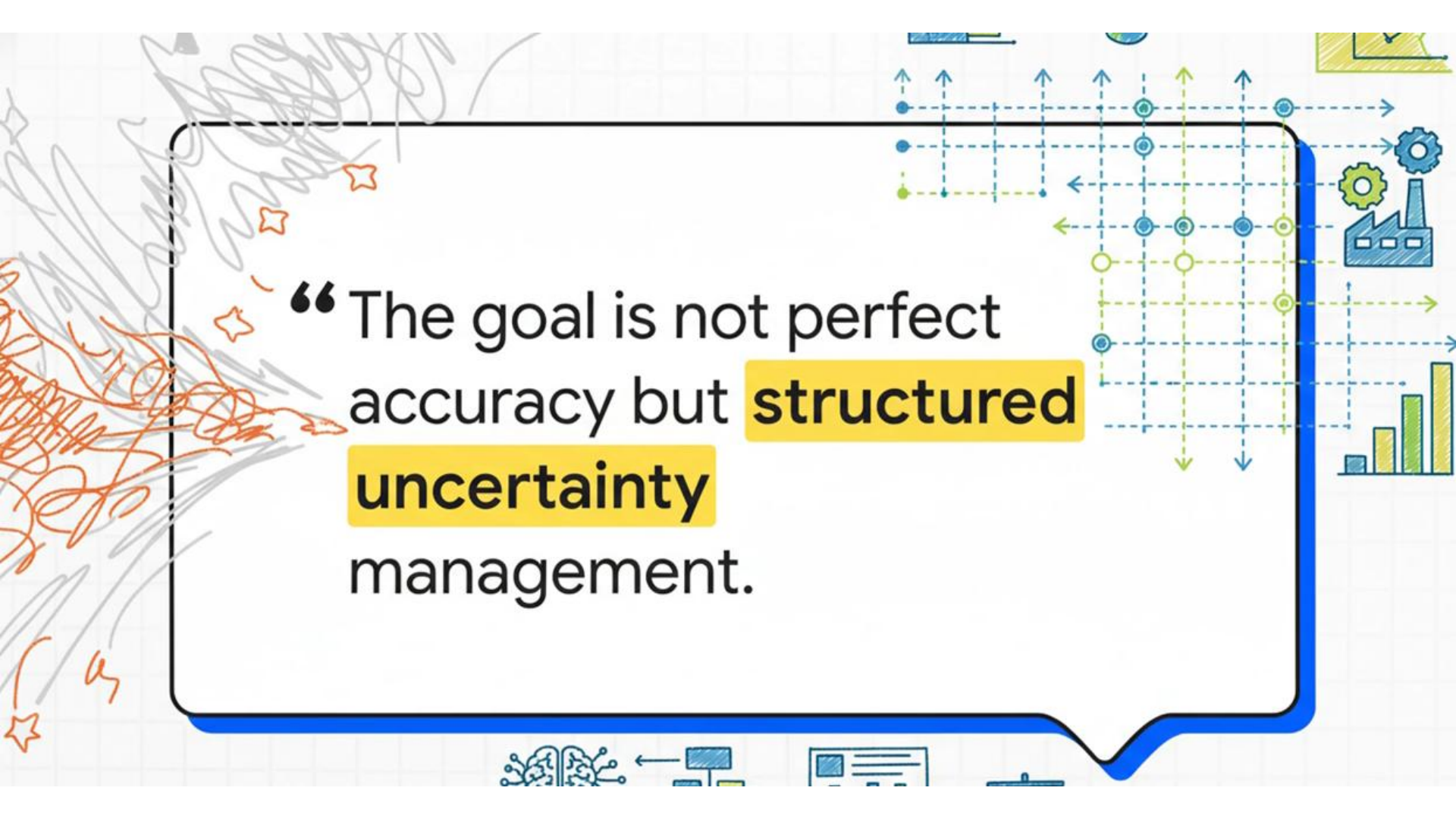
- Planners may distrust model outputs
- Training and pilot programmes essential
- Show wins early to build confidence

3. Model Interpretability

- Black-box models reduce planner trust
- Use explainability tools (e.g. SHAP)
- Governance requires transparent logic

4. Governance Structure

- Define who owns model validation
- Set review cadence in S&OP cycle
- Document decision trails for audit



“The goal is not perfect accuracy but **structured uncertainty** management.

Strategic Takeaway

In project-based rail supply chains, uncertainty cannot be eliminated.
But it can be managed ; and managed well.

01

AI improves forecast quality through multi-signal, nonlinear modelling - not by replacing planners but by augmenting their decisions

02

External signals (infrastructure investment, policy) are as important as internal ERP data in rail demand environments

03

Integration into S&OP governance structures ensures AI outputs become actionable, not just analytical

04

Competitive advantage comes from capital efficiency, backlog reduction, and stronger decision governance - not from automation alone

AI augments planners. It does not replace them.

Discussion & Q&A

